ASSIGNMENT 2

Deadline: 11:59 PM on February 22, 2026

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1 Compulsory Questions

Question 1 (15 %) (Compulsory)

Consider positive and negative rays as $\mathcal{H}$, which is equivalent to the perceptron hypothesis set in 1D. This hypothesis set is often referred to as a "decision stump" due to the shape of its hypotheses.

Specifically, A **positive ray** is defined by a threshold a, where all points $x \geq a$ are labeled $+1$ and all points $x < a$ are labeled -1. Similarly, a **negative ray** is defined by a threshold a, where all points $x \leq a$ are labeled $+1$ and all points $x > a$ are labeled -1. Thus, the hypothesis set $\mathcal{H}$ consists of all possible positive and negative rays defined by different thresholds a.

1. (5%) What is the growth function $m_{\mathcal{H}}(N)$?
2. (10%) What is the minimum break point for $\mathcal{H}$?

Question 2 (20 %) (Compulsory)

Using the VC generalization bound, we know that with probability at least $1-\delta$, the following inequality holds for all $h \in \mathcal{H}$:

$$E_{\text{out}}(h) \leq E_{\text{in}}(h) + \sqrt{\frac{8 \ln\left(\frac{4m_{\mathcal{H}}(2N)}{\delta}\right)}{N}},$$

where $m_{\mathcal{H}}(N)$ is the growth function of $\mathcal{H}$.

1. (10%) Using the Sauer-Shelah lemma, show that the growth function $m_H(2N)$ can be bounded as:

$$m_H(2N) \leq \left(\frac{2eN}{d}\right)^d.$$

   Substitute this bound into the generalization bound to express $E_{\text{out}}(h)$ in terms of $E_{in}(h)$, N, d, and δ.

   Hint: The Sauer-Shelah lemma states that for a hypothesis set H with VC dimension d, the growth function $m_H(N)$ is bounded by:

$$m_H(N) \leq \left(\frac{eN}{d}\right)^d.$$

   The same result can be found at Week 6 slide page 8.

2. (10%) Suppose H has a VC dimension $d = 10$, and you want the generalization error $E_{\text{out}}(h) - E_{\text{in}}(h) \leq 0.1$ with a confidence of 95% ($\delta = 0.05$). Calculate the minimum number of training samples N required to achieve this.

   Hint: You can use online tool like WolframAlpha (https://www.wolframalpha.com/) to sovle N.

Question 3 (15 %) (Compulsory)

Let $\mathcal{H}_{k\text{-int}}$ be the hypothesis set consisting of the union of **at most** k (closed) intervals on the real line. That is, for parameters $a_1 \leq b_1, \ldots, a_k \leq b_k$, define

$$h(x) = \begin{cases} +1, & x \in \bigcup_{j=1}^{k} [a_j, b_j], \\ -1, & \text{otherwise.} \end{cases}$$

(You may allow some of the intervals to be empty, so this class contains all cases with fewer than k intervals as well.)

1. (10%) Show that $\mathcal{H}_{k\text{-int}}$ can shatter $2k$ points on the real line. Provide an argument using the idea of "blocks" of $+1$ labels.

2. (5%) Show that $\mathcal{H}_{k\text{-int}}$ cannot shatter $2k+1$ points. (Hint: consider an alternating labeling $+, -, +, -, \ldots, +$ on $2k+1$ ordered points.) Conclude that the VC dimension $VC(\mathcal{H}_{k\text{-int}})$ is $2k$.

Question 4 (50 %) (Compulsory)

Consider the following simplified learning scenario: assume that the input dimension is one, and the input variable x is uniformly distributed in the interval $[-1, 1]$. The data set consists of 2 points $\{x_1, x_2\}$ and assume that the target function is $f(x) = x^3$. Thus, the full data set is $\mathcal{D} = \{(x_1, x_1^3), (x_2, x_2^3)\}$. The learning algorithm returns the line fitting these two points as g ($\mathcal{H}$ consists of functions of the form $h(x) = ax + b$). We are interested in the test performance (E_{out}) of our learning system with respect to the squared error measure, the bias and the var.

1. (10%) Give the analytic expression for the average function $\bar{g}(x)$.

2. (15%) Write a Python program to simulate the experiment and determine (numerically) $\bar{g}(x)$, E_{out}, bias, and var:

   - Generate $K = 10000$ training sets (each by sampling x_1 and x_2 uniformly from $[-1, 1]$).
   - For each training set $\mathcal{D}_k$, compute $g^{\mathcal{D}_k}(x)$, and numerical squared error measure $E_{out}(g^{\mathcal{D}_k}) = E_x[(g^{\mathcal{D}_k}(x) - f(x))^2]$.
   - Obtain a numerical estimate of $\bar{g}(x)$ as $\bar{g}(x) = \frac{1}{K}\Sigma_{k=1}^{K} g^{\mathcal{D}_k}(x)$.
   - Compute the averaged squared error over all training sets: $E_{out} = \frac{1}{K}\Sigma_{k=1}^{K} E_{out}(g^{\mathcal{D}_k})$
   - Compute at the bias function about x: $bias(x) = (\bar{g}(x) - f(x))^2$, and the variance function: $var(x) = E_{\mathcal{D}}[(g^{\mathcal{D}}(x) - \bar{g}(x))^2]$.
   - Finally, average over x (using the uniform density) to get numerical estimates of bias and variance.
   - Print out the numerical values: E_{out}, bias and var. Produce a plot showing $f(x) = x^3$ along with the average hypothesis $\bar{g}(x)$.

   Attach your code, a screenshot of the print output and the plot in your answer sheet. Then submit an individual code file `q5.py` along with your PDF answer to Blackboard. You may use libraries like `numpy`, `pandas` and `matplotlib` for numerical computations and plotting.

3. (15%) Compare E_{out} with bias+var from your program output. Then compute analytically what E_{out}, bias, and var should be.

Hint: For a uniformly distributed x in $[-1, 1]$, $E[x] = 0, E[x^2] = \frac{1}{3}, E[x^3] = 0, E[x^4] = \frac{1}{5}, E[x^5] = 0, E[x^6] = \frac{1}{7}$.

Question 1

1) $m_H(N) = 2(N+1) - 2$

$= 2N$

2) For break point, $m_H(N) < 2^N$

when $N = 1$, $m_H(N) = 2 = 2^1$

when $N = 2$, $m_H(N) = 4 = 2^2$

When $N = 3$, $m_H(N) = 6 < 2^3$

$\therefore$ min. break point for H is 3.

Question 2

1) Sauer-Shelah lemma:

$$m_H(N) \leq \left(\frac{eN}{d}\right)^d$$

Sub 2N into N,

$$m_H(2N) \leq \left(\frac{e(2N)}{d}\right)^d$$

$$m_H(2N) \leq \left(\frac{2eN}{d}\right)^d \quad \text{—— (1)}$$

Generalization bound:

$$E_{out}(h) \leq E_{in}(h) + \sqrt{\frac{8\ln\left(\frac{4m_H(2N)}{\delta}\right)}{N}} \quad \text{—(2)}$$

Sub (1) into (2),

$$E_{out}(h) \leq E_{in}(h) + \sqrt{\frac{8\ln\left(\frac{4\left(\frac{2eN}{d}\right)^d}{\delta}\right)}{N}}$$

$$E_{out}(h) \leq E_{in}(h) + \sqrt{\frac{8\left(\ln\left(\frac{4}{\delta}\right) + d\ln\left(\frac{2eN}{d}\right)\right)}{N}}$$

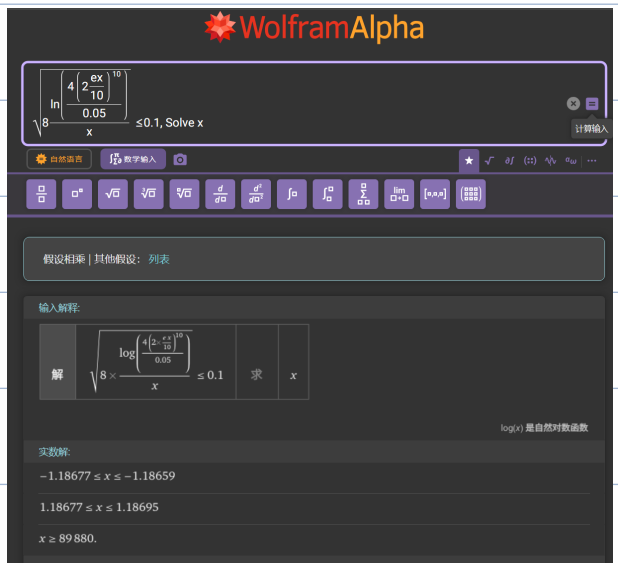
Question 2

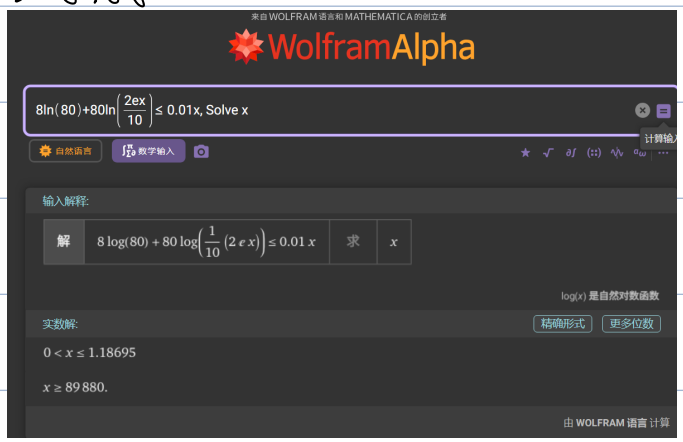
2. $E_{out}(h) - E_{in}(h) \le 0.1$, $\delta = 0.05$, $d = 10$

$$\sqrt{\frac{8\ln\left(\frac{4}{0.05}\right) + 10\ln\left(\frac{2eN}{10}\right)}{N}} \le 0.1$$

$$8\ln 80 + 80\ln\left(\frac{2eN}{10}\right) \le 0.01N$$

$$N \ge 89880$$



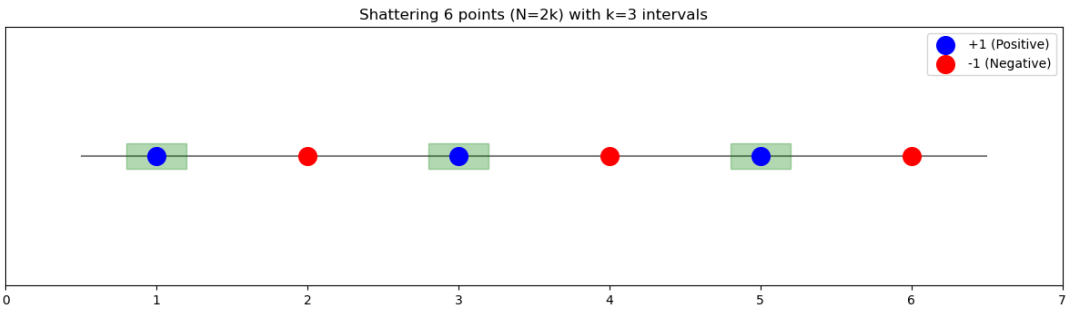


(Change N to x for WolframAlpha to read)

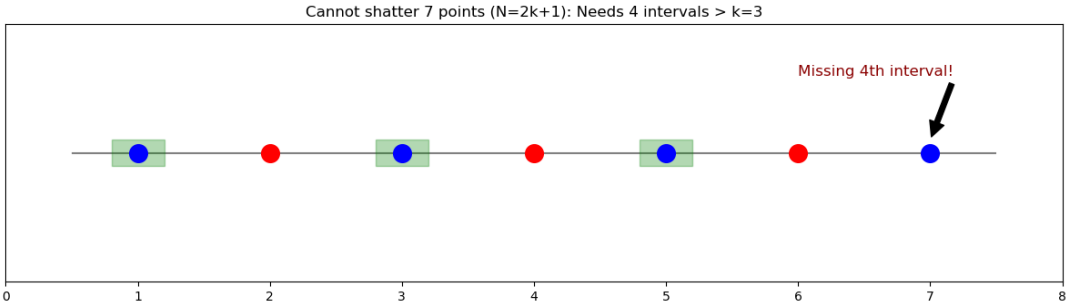
∴ Min. no. of training samples N : 89880

Question 3

1.



2.



1. In the most complex case (+1, −1, +1, −1, ...), there are exactly k blocks of +1 labels. Because we can always cover all +1 blocks with k intervals, $H_{k\text{-int}}$ can shatter 2k points.

2. Consider the most complex case (+1, −1, +1, −1, +1, ...). If we try to cover k+1 positive blocks with only k intervals, ≥1 interval must span across a −1 label, or one +1 block must be left uncovered.

∴ $H_{k\text{-int}}$ cannot shatter 2k+1 points

Question 4

1. target function $f(x) = x^3$

$$D = \{(x_1, x_1^3), (x_2, x_2^3)\}$$

$$a = \frac{x_2^3 - x_1^3}{x_2 - x_1} = \frac{(x_2 - x_1)(x_2^2 + x_2 x_1 + x_1^2)}{x_2 x_1}$$

$$= x_1^2 + x_1 x_2 + x_2^2$$

$$b = x_1^3 - (x_1^2 + x_1 x_2 + x_2^2) x_1$$

$$= -x_1 x_2 (x_1 + x_2)$$

$$E[a] = E[x_1^2] + E[x_1] E[x_2] + E[x_2^2]$$

$$= \frac{1}{3} + 0 + \frac{1}{3}$$

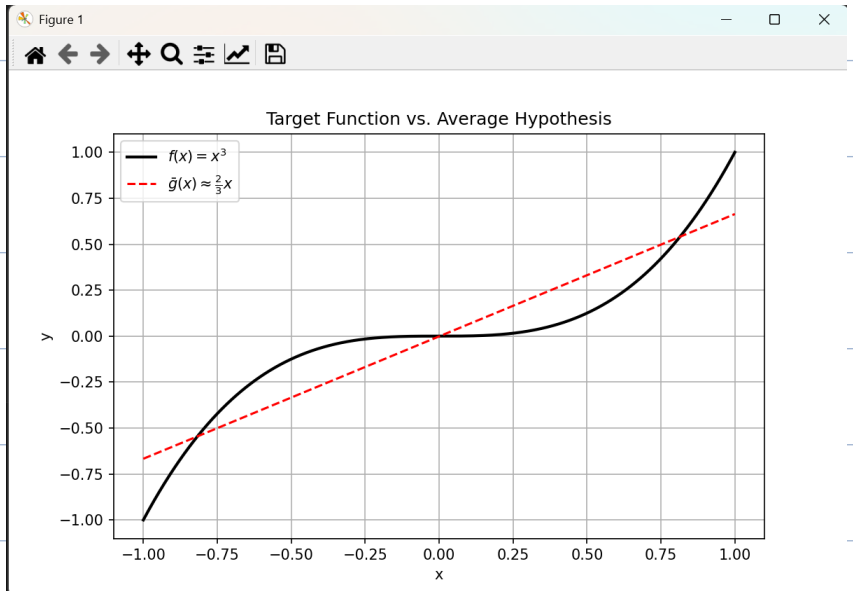
$$= \frac{2}{3}$$

$$E[b] = -E[x_1] E[x_2] (E[x_1] + E[x_2])$$

$$= -0(0)(0 + 0) = 0$$

$$\bar{g}(x) = E[a] + E[b] = \frac{2}{3}$$

2.




```
PS C:\Users\chesn\OneDrive\桌面\CUHK\Year4\Sem2\CSCI3320\Assignment\Assignment 2> python q5.py
Numerical Results (K=10000):
E_out: 0.2496
Bias:  0.0244
Var:   0.2253
Bias + Var: 0.2496
```

Question 4

3. From my program output,

$$E_{out} = 0.2496$$

$$\text{Bias} + \text{Var} = 0.2496 = E_{out}$$

Analytically, $E_{out} = \text{Bias} + \text{Var}$.

$$\text{Bias} = \frac{1}{2}\int_{-1}^{1} (\bar{g}(x) - f(x)\,dx$$

$$= \frac{1}{2}\int_{-1}^{1} \left(\frac{2}{3}x - x^3\right)^2 dx$$

$$= \frac{1}{2}\int_{-1}^{1} \left(\frac{4}{9}x^2 - \frac{4}{3}x^4 + x^6\right) dx$$

$$= \frac{4}{9}E[x^2] - \frac{4}{3}E[x^4] + E[x^6]$$

$$= \frac{4}{9}\left(\frac{1}{3}\right) - \frac{4}{3}\left(\frac{1}{5}\right) + \frac{1}{7}$$

$$= 0.024339$$

$$\text{Var}(x) = E_D\left[(g^D(x) - \bar{g}(x))^2\right]$$

$$= E_D\left[((ax+b) - (\bar{a}x + \bar{b}))^2\right]$$

$$= E_D\left[((a-\bar{a})x + (b-\bar{b}))^2\right]$$

$$E[a^2] = E\left[(x_1^2 + x_1x_2 + x_2^2)^2\right] = 0.524$$

$$\text{Var}(a) = E[a^2] - \left(\frac{2}{3}\right)^2 = 0.08$$

$$\text{Var}(b) = E[b^2] = E\left[x_1^2 x_2^2 (x_1 + x_2)^2\right]$$

$$= \frac{1}{5}\left(\frac{1}{3}\right) + \frac{1}{3} + \frac{1}{5} = 0.133$$

$$\text{Var}(x) = \text{Var}(a) + \text{Var}(b) = 0.213$$

$E_{out_analytically} = 0.213 + 0.024339$

$= 0.2373$

Similar to my program output

Attachment

Declaration for written assignment

I am/we are submitting the assignment for:
☑ an individual project or
□ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged/all members of the group have read and checked that all parts of the piece of work, irrespective of whether they are contributed by individual members or all members as a group, here submitted are original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I/We also acknowledge that I am/we are aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

In the case of a group project, we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic content.

I/we declare that I/we have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I/we declare that I/we have read and understood the University's policy on the use of AI for academic work. I/we confirm that I/we have complied with the instructions given by my/our course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my/our submission.

I/We also understand that assignments without a properly signed declaration by the student concerned and in the case of a group project, by all members of the group concerned, will not be graded by the teacher(s).

Signature(s)	Date 1/2/2025
Name~~(s)~~ Fung Cheuk Nam	Student ID(s) 1155192095
Course code CSCI 3320	Course title Fundamental of Machine Learning

Use of AI tools.

I acknowledge the use of Grok, ChatGPT, Gemini (2026, Feb 22)

for content guidance, code guidance